

3. The diagram below shows an absorption spectrum of an object in space.



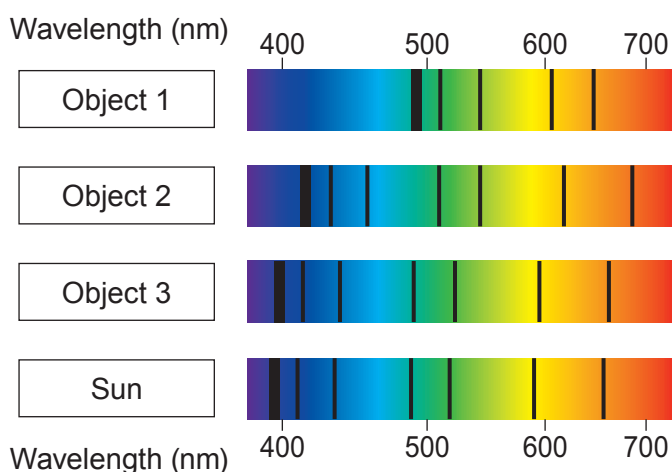
- (a) Complete the following sentences by underlining the correct word(s) in the brackets. [3]

The black lines are caused by light being (**absorbed** / **blocked** / **emitted**) by gases.

The spectrum provides information about the (**size** / **density** / **chemical makeup**) of the object.

The spectrum also provides information about how fast an object is (**expanding** / **using up its fuel** / **moving away from us**).

- (b) The diagram below shows absorption spectra from four different objects in space. The bottom spectrum is from our Sun.



Use the information in the absorption spectra above to answer the following questions. [4]

- (i) State which object is farthest away from Earth.
- (ii) State which object is moving away most slowly from Earth.
- (iii) State **one** conclusion you can make about the chemical makeup of all the objects.
.....
- (iv) State for which object the wavelength of the black lines has increased the most when compared with our Sun.
.....